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Convergent manufacturing of smart metallic structures

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Abstract

Smart Metallic Structures (SMS) featuring embedded sensing capabilities have gained significant attention for mission-critical applications. Despite the growing importance of this emerging field, achieving reliable and efficient fabrication of SMS remains a challenge. This study presents a convergent manufacturing pathway for producing smart structures with integrated sensing capabilities. By leveraging Wire-Arc Additive Manufacturing (WAAM) as the primary additive process and combining it with complementary subtractive machining, sensor embedding, and metallization techniques, the developed approach enables the fabrication of multifunctional metallic components tailored for structural health monitoring (SHM). To demonstrate this concept, a structural beam made of Al-5356 was designed to incorporate a network of strain and temperature sensors. The beam was fabricated using the developed convergent manufacturing approach and subsequently subjected to a series of SHM assessments, including steady-state and transient-thermal analyses, as well as mechanical testing (i.e., three-point bending). The results validate both the structural integrity and sensing functionality of the smart beam prototype, with high-fidelity temperature measurements ($\leq 5\%$ error) and consistent strain response under both the linear and dynamic loading conditions. Despite a moderate mechanical debit reduction ($\sim 21\%$) in flexural modulus and in yield strength ($\sim 19\%$), the beam effectively captured temperature and strain signals throughout testing. The proposed convergent manufacturing approach, together with the demonstrated prototype, holds strong potential for advanced AM and SHM applications.

Keywords: Convergent manufacturing, additive manufacturing; WAAM, smart metallic structures, structural health monitoring (SHM)

1. Introduction

The advent of embedded sensing technologies has enabled the development of smart metallic structures (SMS) that are metal components equipped with integrated sensors capable of capturing key operational data (e.g., strain, temperature, pressure, vibration) in real time [1]. Unlike conventional metallic systems that rely on external instruments or periodic inspections for diagnostics, these intelligent structures offer continuous, non-destructive monitoring, which enhances reliability, supports predictive maintenance, and facilitates early fault detection in mission-critical applications across aerospace, defense, and energy sectors [2].

Recent advances in convergent manufacturing, which integrates additive and subtractive processes within a unified robotic ecosystem, have enhanced the capability to embed sensors into complex geometries with accuracy and mechanical integrity [3]. In particular, metal additive manufacturing has emerged as a powerful platform for embedding sensors within intricate components, combining design freedom and structural robustness [4]. Such advances create opportunities for convergent manufacturing of multifunctional smart metallic parts toward real-time structural health monitoring (SHM).

Nonetheless, the field continues to face key challenges, including: (i) integrating sensors without degrading mechanical properties [5]; (ii) achieving robust bonding between dissimilar materials [6]; (iii) maintaining long-term

sensor functionality under operational loads [7]; and (iv) scaling production for large, complex systems [8]. As such, overcoming these hurdles requires innovative, non-traditional manufacturing solutions.

Building on authors' previous work in this domain [9], this study addresses these challenges through an innovative convergent manufacturing approach that integrates wire-arc additive manufacturing (WAAM) as the primary fabrication method, complemented by subtractive machining, ceramic dispensing, and cold spray metallization for sub-surface sensor embedding. In this approach, a structural beam made of Al-5356 was fabricated comprising embedded network of commercial off-the-shelf (COTS) strain and temperature sensors, serving as the smart metallic structure. The fabricated smart structure (smart beam) is then subjected to a series of structural evaluations, including thermal and mechanical characterizations. This work establishes a convergent manufacturing pathway for producing SMS with sub-surface-embedded sensing capabilities for advanced structural monitoring applications.

2. Design and fabrication of the smart structure

Fig.1a shows the design of the smart structure (beam) (180 mm $\times$ 20 mm $\times$ 5 mm) with COTS strain and temperature sensors embedded beneath the surface. Two strain gauges (SGs, OMEGA, SGT-2/350-XY43) and three resistance temperature detectors (RTDs, OMEGA, F3105)

are strategically positioned within the beam's subsurface for multi-parameter structural monitoring. Representative images of the sensors used in this study are shown in Fig.1b.

As illustrated in Fig.2, the fabrication of the smart beam involves the following sequential steps: (1) WAAM; (2) subtractive machining of sensor cavity followed by sensor bonding; (3) application of ceramic shield (U.S. patent pending); (4) surface metallization via cold spray; and (5) extraction of smart beam and functional integration. The process settings for each step are provided in Table 1.

The fabrication begins with the fabrication of the host metal structure via WAAM using Al-5356 alloyed wire feedstock (Step 1 in Fig.2). Deposition was performed in a sensor-rich WAAM cell [10], equipped with multiple axis robot arm (*Yaskawa*), as shown in Fig.2 (top right panel). Following the building of host structure, sensor cavities were created through subtractive machining using a milling machine (*HAAS CNC*), (Step 2 in Fig. 2). Three cavities,

each measuring 10 mm × 6 mm × 2 mm footprint, were prepared for RTDs, while two slightly larger cavities (14 mm × 6 mm × 2 mm) were dedicated to SGs. The sensors were first mounted on flexible printed circuit board (FPCB) via offline spot welding to facilitate electrical connections and mechanical handling. The packaged sensor assemblies were then bonded into the machined cavities using a dedicated adhesive (*M-Bond 610*).

Table 1. Process settings for each manufacturing step.

Fabrication Process	Parameter	Value (unit)
Step-1 WAAM	Material	Aluminum 5356
	Wire feed rate	~ 0.17 m/s
	Torch speed	4 mm/sec
	Weld beads Number	20
	Beads distance	6 mm
Step-2 Machining	Cutting tool	3.175 mm end mill
	Spindle speed	4,000 rpm
	Feed rate	~ 50 mm/min
	Material	Al, Al ₂ O ₃ , Zn
Step-3 Surface metallization	Gas temperature	400 °C
	Gas pressure	6 bar (0.6 MPa)
	Nozzle transverse speed	20 mm/sec
	Nozzle stand-off distance	25 mm

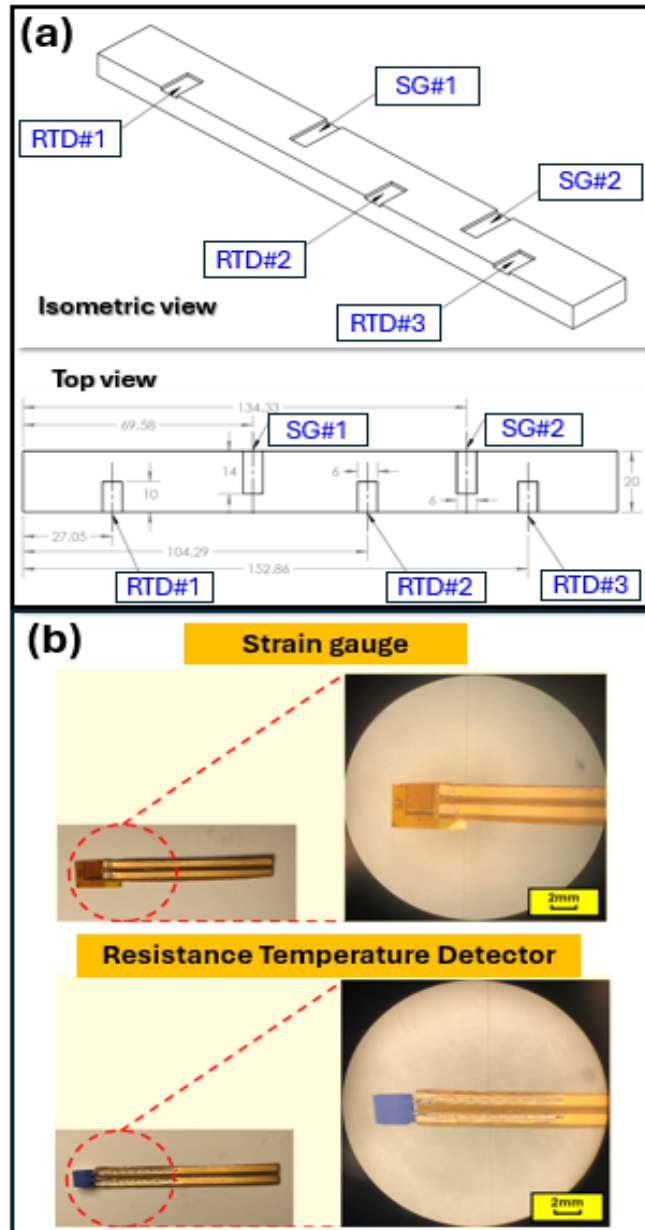


Fig. 1. (a) Smart beam design with sub-surface embedded sensors; (b) COTS sensors embedded into the beam.

To protect the embedded sensors from processing-related hazards and environmental stressors, a ceramic layer was applied over the FPCB-packed sensors, followed by 24 hours of curing at room temperature (see Step 3 in Fig.2). After curing, the ceramic layer remained in the green phase, retaining sufficient ductility to support the subsequent CSAM process. As such, this ceramic shielding serves as a protective barrier during CSAM, as well as against potential moisture, oxidation, and abrasion.

Subsequently, the ceramic surface was metallized and encapsulated using CSAM (see Step 4 in Fig. 2). Herein, CSAM enables solid-state AM directly onto the ceramic surface, achieving seamless metallization of the sensor cavity while providing mechanical reinforcement and enhanced thermal stability for the embedded sensors. Furthermore, the solid-state nature of CSAM minimizes thermal loading [11], thereby preserving the integrity and calibration of the sensing elements during fabrication.

Lastly, sensor leads were soldered with a tin-lead alloy to connect the beam to a data acquisition unit (*NI DAQexpress*), (Step 5 in Fig.2). The beam was then subjected to SHM evaluations as described in the following sections.

3. Performance evaluation of the smart structure

In this section, the fabricated smart beam is evaluated through structural tests comprising; (i) steady-state thermal analysis; (ii) transient thermal analysis; and (iii) three-point bending tests. These tests assessed the beam's real-time sensing performance under varying thermal and mechanical loads. Detailed procedures for each test are described below.

3.1. Sensor calibration and steady-state thermal analysis

The RTD sensors were first calibrated under steady-state thermal conditions in a thermal chamber (Fig.3a) following the IEC-60751 standard [12]. Four surface-mounted thermocouples (TCs) served as reference measurements (i.e.,

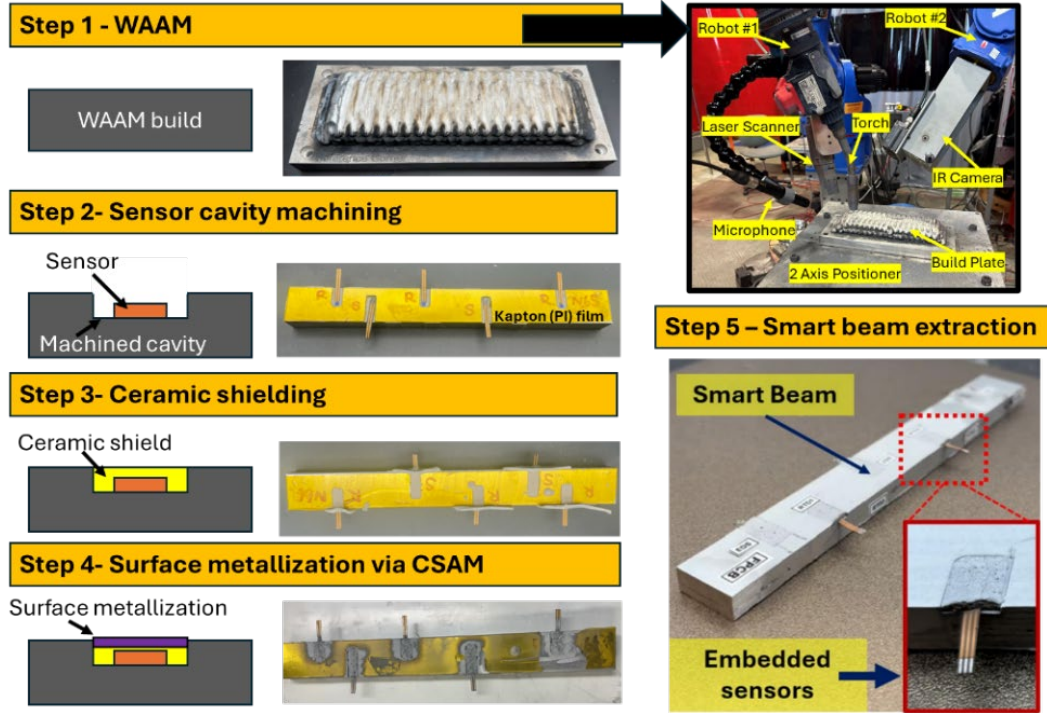


Fig. 2. Convergent manufacturing pathway for smart metallic structures.

ground truth) for the embedded RTDs (Fig.3b). Steady-state data were collected at 60 °C, as shown in Fig.3c. Given the narrow temperature range and negligible quadratic term, the calibration model was simplified to a linear form, which achieved relative errors below 0.5% (Figs. 3c-d). These results confirmed that embedded RTDs can accurately measure subsurface temperatures, supporting applications in thermal characterization and real-time structural diagnosis.

3.2 Transient thermal analysis

As shown in Figs. 3d, a transient heating experiment was conducted by applying a programmed thermal stimulus to the beam's leftmost surface using a controlled ceramic heater. Surface TCs and embedded RTDs were used for cross-validation. RTD 1, located near the heat source, exhibited a slight response delay due to thermal conduction from the heated surface to the embedded sensor location. However, this delay was considered negligible, as no reference thermocouple was available at that position and RTD 1 was primarily used to represent the heat input. Despite the delay, RTD 1 accurately captured the overall surface temperature trend (see Fig. 3e). Sensors positioned farther from the source recorded smaller temperature variations yet maintained consistent temperature evolution, as shown in Fig. 3e. As such, the transient-thermal tests confirmed that embedded RTDs provide accurate temperature measurements across varying distances from the heat source, validating their capability for real-time monitoring of dynamic thermal conditions in structural applications.

3.3 Mechanical testing

Three-point bending tests were conducted to (i) verify the functionality of embedded SGs and (ii) compare the mechanical properties of the smart beam to those of baseline beam (i.e., pristine beam without embedded sensors). In the elastic-range test (1.067 mm/min to 0.5 mm displacement, 2 min hold), two surface-mounted SGs served as reference

measurements (Fig. 4a). All SGs accurately tracked load variations (Fig. 4b), with subsurface SGs recording compressive strains and surface SGs tensile strains, consistent with bending theory [13]. Sensors positioned closer to the center exhibited larger strains, further confirming the accuracy of the embedded SGs.

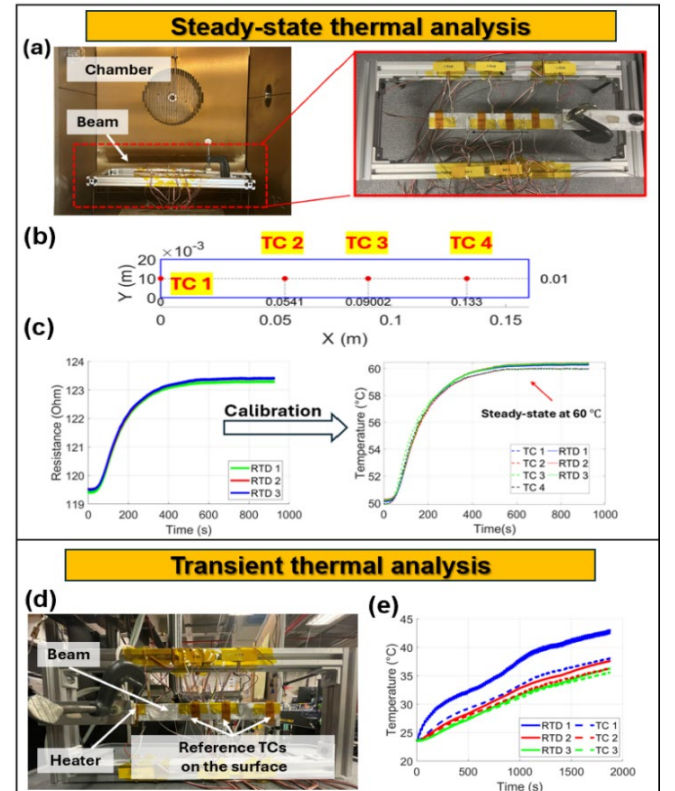


Fig. 3. (a) Steady-state thermal experiment setup; (b) Placement of surface-mounted reference TCs along the beam; (c) RTD calibration (left) and thermal measurements (right); and (d) transient thermal responses of embedded RTDs and TCs.

A follow-up displacement-controlled test (1.846 mm/min to failure) was conducted to estimate the flexural modulus, yield strength, failure strain, and maximum stress (Fig. 4c). Compared with the baseline beam, the smart beam exhibited nominal reductions of 20.6% in flexural modulus and 18.9% in yield strength. These results suggest that the embedded sensing structure may introduce a moderate reduction in stiffness and strength. The failure strain exhibited only a slight change compared to the baseline beam. Within the resolution of the present preliminary tests, the smart beam largely maintained its deformation capacity.

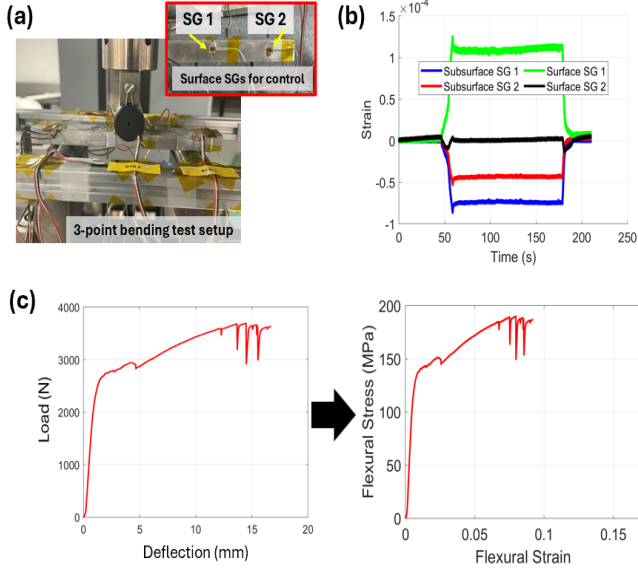


Fig. 4. Mechanical structural monitoring experiments: (a) Three-point bending test setup for mechanical characterizations; (b) Strain time history within the elastic range; (c) Load–deflection response (left panel) and flexural stress–strain curve (right panel).

Table 2. Comparative summary of mechanical performance metrics.

Property	Baseline beam	Smart beam	Mechanical debit (%)
Flexural modulus (GPa)	66.12	52.49	20.6
Yield strength (MPa)	148.24	120.4	18.9
Failure strain	0.077	0.075	2.6

4. Conclusion

This study presented a convergent manufacturing pathway for SMSs with sub-surface embedded sensing capabilities. A smart beam equipped with sub-surface temperature and strain sensors was fabricated using a hybrid additive-subtractive manufacturing process. The key takeaways include:

- RTDs delivered stable, high-accuracy temperature readings under both steady-state and transient conditions, with an error margin below 0.5%.
- SGs provided high-fidelity strain measurements during both elastic and failure-range loadings.
- The smart structure exhibited moderate reductions in flexural modulus (20.6%), yield strength (18.9%), and failure strain (3.6%) compared to the baseline beam, yet maintained strong structural performance for embedded sensing.

- The proposed convergent manufacturing approach demonstrated the viability of integrating sensing into structural metallic components for effective SHM.

Future work will focus on resuming additive manufacturing over sub-surface embedded sensor network to achieve fully embedded sensor network for mission critical SHM applications in aerospace and defence.

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